

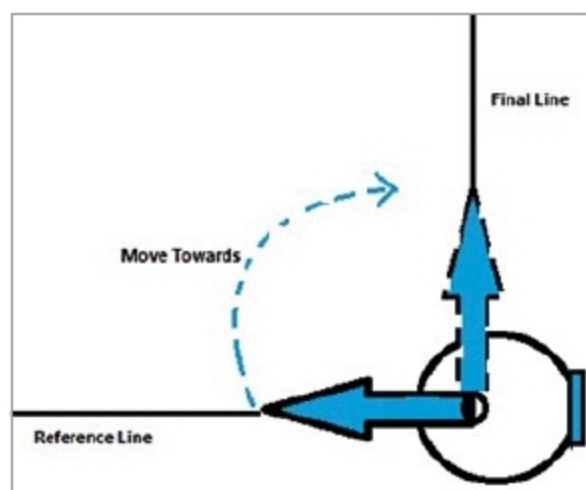


Fig. 8: Principle of protractor needle movement

EFY Note
The source code
of this project
is available
for free download
at source.efymag.com

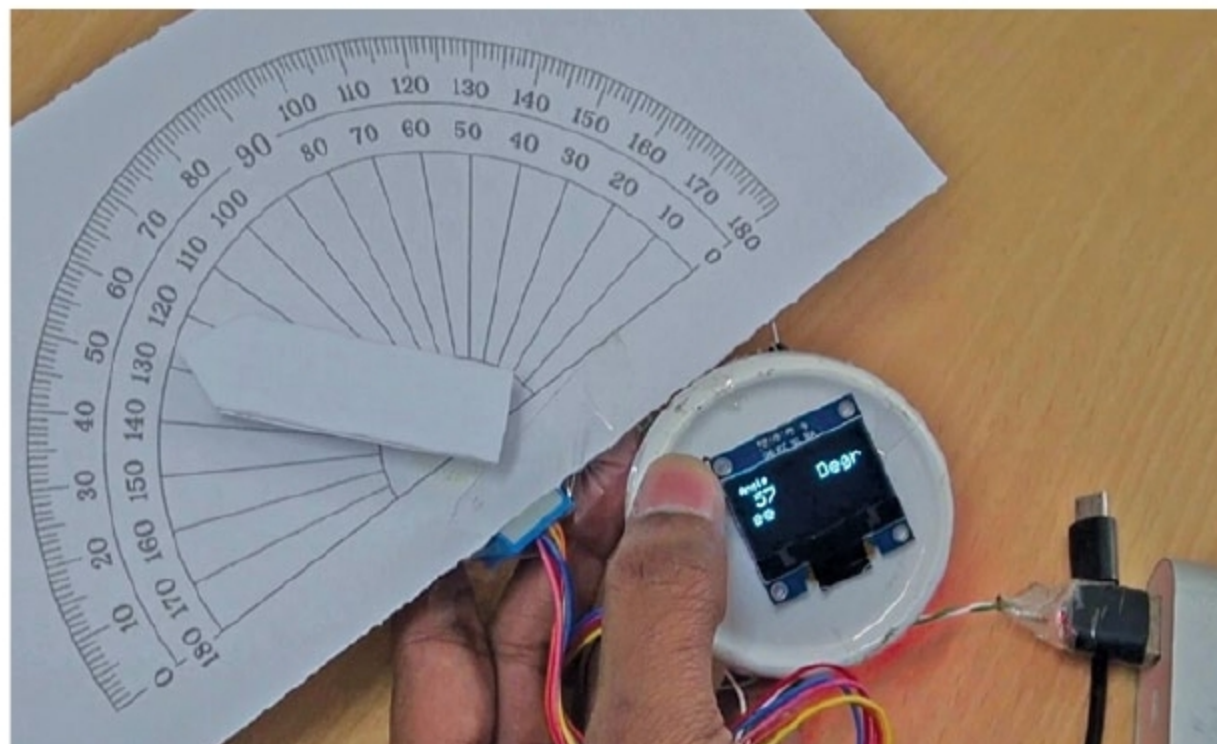


Fig. 9: Digital protractor calibration

tractor as shown in Fig. 7. The principle of protractor needle movement is shown in Fig. 8.

The digital protractor can be calibrated by using a protractor mapped on a piece of paper as shown in Fig. 9. A fixed needle or pointer is to be attached on the stepper motor parallel to 180-degree angle line of the protractor, which acts as reference line. This needle can be placed on the bottom side of the protractor during calibration. A moveable needle is to be attached to the shaft of the stepper motor that is used for angle measurement.

The moveable needle can move in clockwise or anti-clockwise direction with respect to the fixed reference needle. The angle between the reference needle and moveable needle is displayed on the OLED display.

When the moveable needle rests on the reference line on the left side, the angle read is 0. When you press button S1, the moveable needle moves in

clockwise direction until it reaches the desired position (the final line of angle).

Testing

Double check all the circuit connections, as any mistake might fry your costly components. Then power on the Arduino board and ULN2003A motor driver module. Press switch S1. As the needle starts moving clockwise, the angle moved with respect to the reference line is shown on the OLED display. If you press switch S2, the stepper motor will start moving in opposite direction, and angle value shall start decreasing, which can be seen on the OLED display.

This protractor is more useful for measuring the angle between adjacent walls and solid structures, links and joints as it is very hard to put a conventional measuring device in such places. **EFY**

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